

Autonomous Suturing Task: From Hierarchical Policies in Simulation to a Guarded Needle-Handling Pipeline on a Real dVRK

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Abstract—Reinforcement-learning (RL) policies for surgical suturing are commonly trained and scored in simulation and seldom run on a physical robot. We report our progress in moving the hierarchical suturing policies of SurgicAI onto a da Vinci Research Kit (dVRK). First, we re-run the original five-stage hierarchy (Approach, Place, Insert, Regrasp, Pullout) on ROS 2 and AMBF 3 without changing any policy or environment logic. It completes the full suture in 11 of 20 episodes, against the 52% reported by its authors. Second, we build a perception front end for the real endoscope. Depth Anything V2 is fine-tuned on simulated RGB-depth pairs so that it outputs metric depth, and FoundationPose uses that depth to estimate the pose of the needle. An ArUco-based hand-eye calibration improves PSM localization from about 5 cm to 5–10 mm. Third, we deploy the released policies on a real PSM through an explicit observation-action contract. Running the policies against their own demonstrations exposed three silent mismatches between the training environment and a straightforward deployment: the Euler-angle branch, the action scale, and whether the observation is open- or closed-loop. Once these are fixed, the released Approach checkpoint reproduces 45 of its 50 demonstrations. We also show that the gap between the training workspace and the real task can be closed by staging the start pose rather than by retraining, and we describe a guarded phase machine that takes the arm from staging through grasp, lift and transport to the suture entry pose. Finally, we propose a two-stage grasp calibration, in which a Bernstein residual field corrects a coarse FoundationPose-based grasp target, and report the first ten hardware samples. They reveal a constant offset of about 43 mm that must be removed before any fit.

I. INTRODUCTION

Suturing is the canonical benchmark for surgical autonomy. It is long-horizon, contact-rich, and needs millimetre precision on a thin, curved, near-symmetric object. Simulation platforms built on AMBF [1] and the Surgical Robotics Challenge (SRC) [2] have made the task learnable. SurgicAI [3] splits it into five subtasks and learns a policy for each one. A high-level policy then sequences them, and the full chain succeeds far more often with this hierarchy than with a flat policy. Physical deployments remain rare. Orbit-Surgical

transferred a policy to a real dVRK at 50% on needle lift [4], and a recent review describes the field as “mature at the simulation stage but ... preclinical” [5].

This draft reports what it has taken so far to move the SurgicAI policies onto a real dVRK [6] Patient Side Manipulator (PSM). We emphasize where the transfer actually broke. Much of the gap turned out not to be “sim-to-real” in the usual sense. The cause was usually an unstated convention in the training code that a deployment has to reproduce exactly. Our contributions are:

- 1) A *replication* of the hierarchical suturing pipeline on ROS 2/AMBF 3. The environment and policy code are unchanged, and the chained success rates are consistent with the published ones (Sec. III).
- 2) A *perception front end* for the real endoscope: supervised metric depth, FoundationPose needle tracking and hand-eye registration of the PSM (Sec. IV).
- 3) A *verified deployment contract* for the released checkpoints. It identifies three defects that invalidate a straightforward deployment, and a staging solve that places any real grasp inside the trained support (Sec. V).
- 4) A *guarded real-robot phase machine* and the hardware findings from bringing it up, including the PSM position deadband (Sec. VI).
- 5) A *two-stage grasp calibration* whose structure is predicted from the dVRK kinematics before any data is collected, together with the first hardware samples (Sec. VII).

Fig. 1 shows how the components fit together.

II. RELATED WORK

Learning suturing in simulation. AMBF [1] and the SRC [2] provide a dVRK-faithful suturing scene with a needle, a phantom and entry/exit targets. SurgicAI [3] benchmarks RL and imitation-learning algorithms on its subtasks. TD3 [7] with hindsight experience replay [8] and a behavior-cloning term reaches roughly 96% on single subtasks. The hierarchical composition reaches 52% on the full chain, compared with 15% for a flat policy. Orbit-Surgical [4] trains in GPU-parallel simulation and reports a physical transfer.

Sim-to-real. Domain randomization is the principal mitigation. Xie *et al.* [9] find that randomizing one dimension at a time transfers more reliably than randomizing everything at once. We adopt their stopping rule for a retraining curriculum, but our results suggest that the first-order transfer

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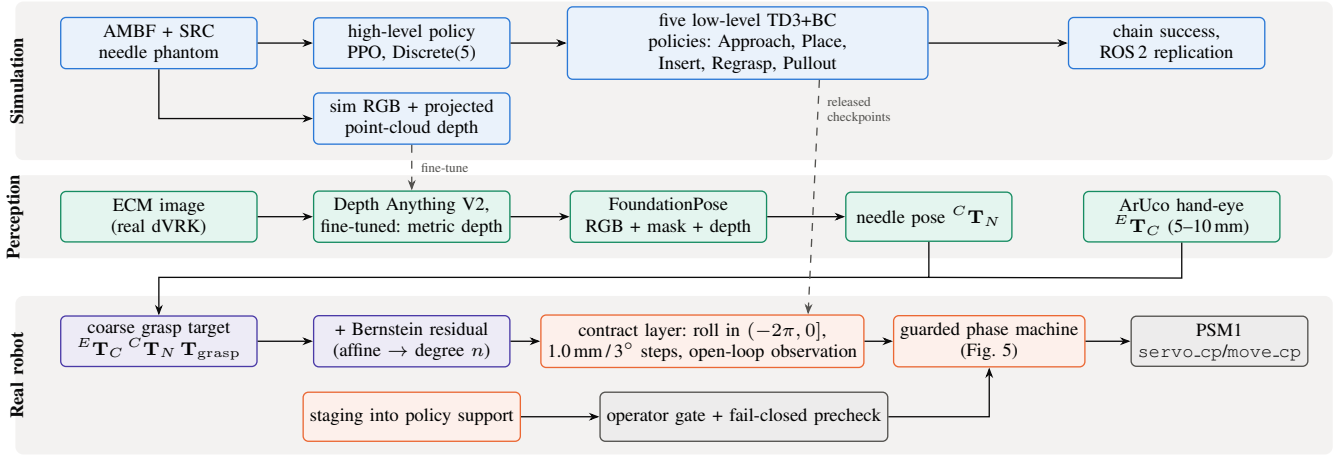


Fig. 1. System overview. *Simulation*: the SurgicAI hierarchy in AMBF/SRC, re-run on ROS 2; the same simulator provides RGB–depth pairs for depth fine-tuning. *Perception*: on the real dVRK, fine-tuned Depth Anything V2 supplies metric depth to FoundationPose, and an ArUco hand-eye calibration registers the camera to the PSM. *Real robot*: the perceived needle pose becomes a coarse grasp target, which a learned residual corrects. The released checkpoints drive the arm only through a verified contract layer and a guarded phase machine.

TABLE I
STAGES OF THE HIERARCHY, AS USED BY THE LOW-LEVEL CONTROLLER.

Task	Stage	Arm	Completion side effect
1	Approach	PSM2	close jaw, attach needle
2	Place	PSM2	re-sync goal from measured pose
3	Insert	PSM2	–
4	Regrasp	PSM1	PSM1 grasps, PSM2 opens
5	Pullout	PSM1	episode terminates with success

problems for these policies are contract mismatches rather than visual or dynamic gaps.

Needle pose and robot calibration. Markerless needle pose tracking reaches roughly 1 mm [10]. FoundationPose [11] provides model-based 6D pose from RGB-D with no object-specific training. Kinematic error on the dVRK is dominated by joint offsets and cable effects [12]. Hwang *et al.* need a recurrent model because hysteresis is not a function of position. We build on the two-stage calibration of Xiao *et al.* [13], a coarse geometric model followed by a Bernstein-polynomial residual.

III. HIERARCHICAL SUTURING IN SIMULATION

A. The SurgicAI hierarchy

The hierarchy has a high-level PPO policy with a 35-dimensional observation and five discrete actions. Each action selects a low-level TD3+BC policy (Table I), and each low-level policy runs for at most 200 steps. Every low-level policy is goal-conditioned on a 7-DoF target made of position, roll–pitch–yaw and jaw. It outputs a normalized increment that is scaled by a per-task step size and integrated into the commanded pose.

B. Porting to ROS 2 and AMBF 3

The original code targets ROS 1 Noetic, AMBF 2.0 and a 2024 snapshot of the SRC. We run it unchanged on

Ubuntu 24.04 with ROS 2 Jazzy by adding three pieces. The first is a minimal `rospy` shim built on the ROS abstraction layer that the ROS 2 AMBF client already uses. The second is a set of repository-relative model and config paths. The third is a set of runtime fixes for three problems in the ROS 2 AMBF client that only the old scene exposes: sensors that do not accept the middleware handle, incremental DDS discovery that silently misses objects, and name lookups made ambiguous by ghost objects. Checkpoints saved under Python 3.8 load under Python 3.12 with Stable-Baselines3 2.2.1 and give identical predictions once training-only pickles are skipped.

C. Results

The high-level PPO checkpoint has not been released. We therefore sequence the five stages in order and allow up to two retries per stage. This stands in for a high-level policy that keeps selecting a stage until it completes. The low-level checkpoints are recovered from the repository history. Fig. 2 compares 20 episodes (seed 10) with Table 3 of [3] (100 episodes; “Regrasp” is called “Handoff” there). The full chain succeeds in 11/20 episodes (0.55), against 0.52 in the paper. All nine failures reach the 800-step low-level limit: seven in Pullout and two in Place. The mean number of steps per completed stage is 139 for Approach, 59 for Place, 88 for Insert, 154 for Regrasp and 111 for Pullout. We also found a bookkeeping bug. `info["is_success"]` is set when Pullout completes but is overwritten later in the same function, so it is never true. Success is therefore counted from `terminated`, which is also what the high-level reward uses.

The intermediate chains exceed the published values (0.90 against 0.72 and 0.60). The likeliest reason is our stage sequencing: it always retries rather than letting a learned high-level policy choose, and a retry can recover a stage that a policy would have abandoned. Table II shows why the policies cannot run on the newer SRC scenes without

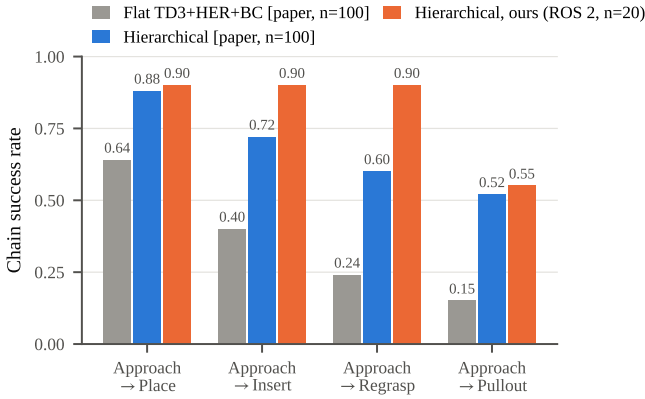


Fig. 2. Chained success from Approach to each later stage. Grey and blue: flat and hierarchical results from [3] ($n=100$); the flat TD3+HER baseline, 0.00 throughout, is omitted. Orange: our replication on ROS 2/AMBF 3 ($n=20$), with scripted stage sequencing in place of the unreleased high-level policy. With $n=20$ the 95% interval on 0.55 is about ± 0.22 .

TABLE II

WHAT THE POLICIES ASSUME, OLD VS. CURRENT SRC (STATIC CHECK).

Assumption	Old SRC (policies)	SRC v2.x
middleware	ROS 1	ROS 2
needle topic	env/Needle	env/phantom/Needle
FK module	psmFK	removed
PSM1 base	(1.0, 1.346, 1.350), 0.1-scaled	(0.131, 0.397, 0.878) m
needle spawn	hard-coded world pose	differs per phantom

adaptation. The observations are absolute poses in the PSM base frame, and the base-to-needle offset changes by several centimetres between the old and new worlds.

IV. PERCEPTION ON THE REAL dVRK

A. Simulation-Based Adaptation of Depth Anything for Metric Depth Estimation

We first evaluated the pretrained Depth Anything V2 model [14] on images from the simulated suturing environment without task-specific fine-tuning. In this baseline configuration the model produced relative depth estimates, normalized to a range of approximately 0 to 1, rather than physically meaningful metric depth. The pretrained model had also never seen the visual characteristics of the surgical scene: the surgical instruments, the tissue phantom and the endoscopic imaging conditions. Its depth predictions therefore had substantial errors and were not accurate enough for downstream robotic perception and manipulation.

To adapt the model to the surgical domain and make it predict absolute depth in meters, we fine-tuned it on data generated from the AMBF-based suturing simulation environment. Collecting equivalent supervised training data directly from the physical dVRK is difficult because the real endoscopic camera provides no dense ground-truth depth. The simulator is therefore an effective source of paired RGB and metric-depth data that preserves the visual and geometric characteristics of the target surgical scenario.

During data collection, RGB images were acquired from the simulated endoscopic camera and used as inputs to the depth estimation network. For each RGB frame, the AMBF simulator also provided a corresponding three-dimensional point cloud. The 3-D points were transformed into the camera coordinate system and projected onto the image plane to build a dense per-pixel depth map in meters, which was used as ground-truth supervision for fine-tuning. Each training sample was thus an RGB image of the simulated suturing scene paired with its geometrically derived metric-depth map.

After task-specific fine-tuning, the model estimated depth substantially better within the suturing environment. Instead of relative depth, the adapted model predicts depth directly in metric units and reproduces the spatial structure of the surgical scene more accurately. This simulation-based training strategy is a practical way to obtain dense depth supervision that is difficult to acquire on the physical dVRK, and it adapts the pretrained depth estimator to the geometric and visual characteristics of robot-assisted suturing. [Jiaming: add the quantitative depth error before and after fine-tuning (e.g. AbsRel, RMSE in mm) and the size of the training set.]

B. FoundationPose Tracking on the Real dVRK System

[unfinished – Jiaming: add experimental results and explain the observed tracking failure.] FoundationPose [11] was used to estimate the poses of the PSM and the surgical needle on the physical dVRK. The method requires an RGB image, an object mask and a depth map as inputs. Because the dVRK endoscope provides no direct depth measurements, the depth maps were generated with the fine-tuned Depth Anything model.

The pipeline tracked the surgical needle reliably. Tracking of the PSM was less stable and had larger pose errors. When tracking the PSM [...]

In simulation, the complete path from RGB through Depth Anything, FoundationPose and a first-frame physical-plausibility gate to a frozen needle goal passed the gate on 40/40 captures. A Reach controller driven by these goals succeeded in 29/30 episodes. [Confirm which goal source (GT vs. FP) the 29/30 refers to; these are historical reference runs and were not re-run for this draft.] Needle yaw in the curriculum is capped at $\pm 30^\circ$ because the pose audit was reliable at 20° but produced near- 180° flips at 40° . Section V-C shows that this cap, not the policy’s trained workspace, is what limits the task.

C. Hand-Eye Calibration for PSM Localization

To obtain an independent estimate of the PSM pose in the camera frame, we performed hand-eye calibration with the dVRK camera registration framework. An ArUco marker [15] was mounted on the PSM shaft approximately 70 mm from the instrument tip. The camera was positioned so that the marker stayed near the center of the image, and the PSM was moved through multiple poses while the marker remained visible. The recorded robot and marker poses were

then used to estimate the transformation between the PSM and camera frames.

This calibration improves the PSM localization accuracy from approximately 5 cm to 5–10 mm. The resulting pose estimate is an additional reference that can be fused with the FoundationPose estimate, for example with a Kalman filter, to make PSM tracking more robust. Section VII explains why hand-eye error of this size is exactly affine in position and can therefore be absorbed by the grasp calibration.

V. DEPLOYING THE RELEASED POLICIES

We deploy three checkpoints: the released SurgicAI Approach and Place policies (TD3+HER+BC) and R6, a locally retrained single-goal Approach policy. Deployment is inference-only. It uses a CPU-loaded 3×256 actor and the arm’s `measured_cp` in the ECM frame. It does not use AMBF or any training code.

A. The observation–action contract

The contract is decoded from the checkpoints themselves.

Observation: $o = [a, d, d-a] \in \mathbb{R}^{21}$, where a and d are the achieved and desired poses. Each is written as position in centimetres, RPY in radians (extrinsic xyz , KDL `GetRPY`), and jaw normalized to $[0, 1]$. Our observation builder reproduces all 502 observations stored in the R6 checkpoint exactly.

Action: $u \in [-1, 1]^7$ is integrated as $c_{t+1} = c_t + u \odot s$. The step s is in *metres* and radians while the observation is in centimetres, and this asymmetry is part of the contract.

B. Three silent defects

A deployment can satisfy all of the above and still fail. We replayed each checkpoint from its own demonstration starts against a kinematic arm model, and this found three further defects. Together they invalidate every hardware RL comparison we had made before fixing them.

1) *The roll branch.* A rotation matrix is branch-invariant, but its RPY triple is not. SciPy returns roll in $[-\pi, \pi]$, whereas SurgicAI’s `Frame2Vec` maps it into $(-2\pi, 0]$. All stored goal rolls in the released checkpoints lie outside $\pm\pi$, and none lies outside $(-2\pi, 0]$. With the SciPy convention, upstream Approach reproduces 0/25 demonstrations. The geometric servo was never affected because it uses only relative rotation. This is why “the servo works and RL does not” looked like a statement about policy quality for so long.

2) *Two action scales.* The environment metadata stores $0.5 \text{ mm}/2^\circ$, the scale the *demonstrations* integrate at, and fitting the demonstrations recovers this value exactly. The training and evaluation scripts, however, hard-code $1.0 \text{ mm}/3^\circ$, and all published success rates were measured at that scale. Replay picks $1.0 \text{ mm}/3^\circ$ unambiguously (Fig. 3a). The 1.5 mm we had been applying came from a utility module and was never checked against the policy.

3) *Open- vs. closed-loop observation.* Inside a subtask, SurgicAI feeds the network the integrated *command*, never the measured pose. A perfect arm hides the difference. For a lagging arm, feeding the measured pose drops Approach

from 25/25 to 6/25 at 30% tracking per cycle (Fig. 3b). We now compute the observation from an internal integrator seeded once, and the measured pose is used only by the guards. A fourth defect appeared while testing this fix. The per-step safety caps were measured against the measured pose, so they pulled the command back toward a lagging arm on every cycle. This re-closed the loop from inside the safety layer. The caps now bound consecutive commands, and a separate guard bounds tracking error.

With all three defects fixed, upstream Approach reproduces 45/50 of its demonstrations through our loop (published $96 \pm 6\%$), and R6 reproduces 46/50. We also found that the published rates use a tolerance of $0.5 \text{ cm}/30^\circ$. At that tolerance Approach reproduces 25/25 and Place 22/25. At 10° , the tolerance in the environment metadata, Place drops to 12/25, so Place is not a precision orientation controller.

C. The workspace question is a staging question

On the real task, the released Approach policy is out of distribution. The needed start-to-goal rotation is $0\text{--}30^\circ$, while training covered $53\text{--}97^\circ$, and the tool-frame offset has the wrong sign in y . Every unstaged RL run orbited the goal. We first expected this to require retraining on a wider workspace. However, the demonstrated support of a goal-conditioned policy is expressed entirely in relative quantities:

$$\delta = R_s^\top (p_g - p_s), \quad \theta = \angle(R_s, R_g), \quad (1)$$

and both are invariant to rigid transforms. For any grasp pose (p_g, R_g) there is therefore a start pose inside the support, and it can be solved for directly:

$$R_s = R_g \text{Rel}^{-1}, \quad p_s = p_g - R_s \delta^*, \quad (2)$$

where `Rel` and δ^* are the rotation and offset taken from the support. We evaluate this on the recorded geometry of our dVRK with 400 needle placements over a $\pm 3 \text{ mm}$, $\pm 30^\circ$ envelope. Without staging, 0/400 placements are in support. With staging, all 400 are, and the staging move has a median length of 3.66 cm and a maximum of 4.14 cm (Fig. 4). For the approach leg, the trained workspace is not the binding constraint. The binding constraints are perception (the $\pm 30^\circ$ yaw cap), reachability and Place.

D. The trained goal is 7 mm short of the needle

SurgicAI builds the Approach goal as $\mathbf{T}_N(\phi) \text{Trans}(0, 0, 7 \text{ mm}) R_{\text{flip}}$. It is the grasp point on the arc, backed off 7 mm along the tool axis with the jaws pointing at it. In simulation the grasp is then faked by attaching the needle wherever the jaws are, so no trained motion ever covers the last 7 mm. On hardware, closing the jaw at the policy goal would close on empty space. We therefore add a *descend* phase that covers the standoff at 0.5 mm per cycle along the jaw axis with the orientation frozen. It is the slowest motion in the sequence because it is the only one that can disturb the needle before the needle is held.

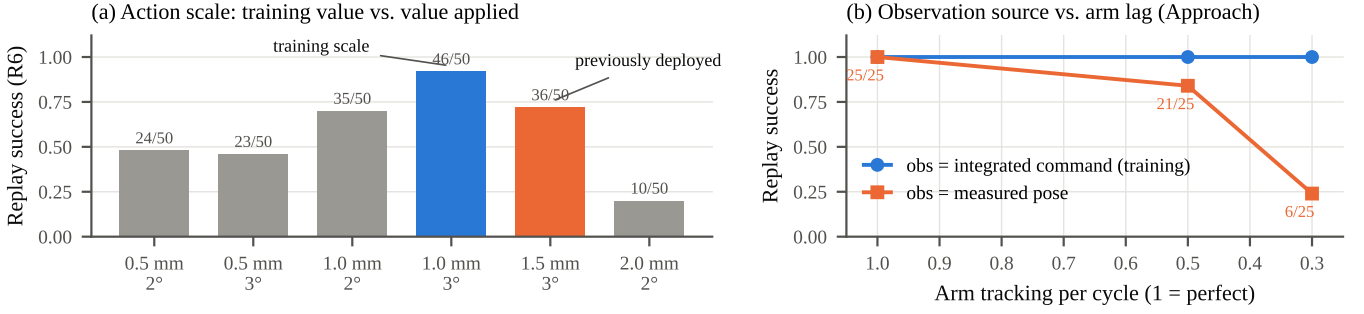


Fig. 3. Two contract defects, measured by replaying policies from their own demonstration starts. (a) R6 success as a function of action scale at its own tolerance (1 cm/10°, 200 steps): the training scale gives 46/50, and the scale previously deployed gives 36/50. (b) Upstream Approach under a first-order lagging arm. Feeding the measured pose instead of the integrated command matches the training environment only for a perfect arm.

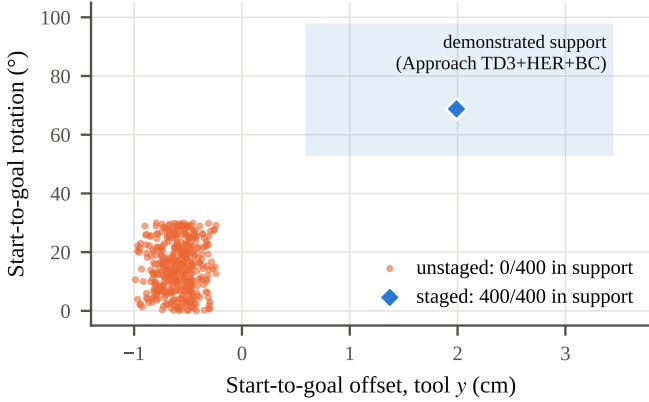


Fig. 4. Relative start-to-goal geometry for 400 needle placements on the recorded dVRK task, against the demonstrated support of the upstream Approach policy. Staging moves every placement to the same point inside the support. It is a 3.7 cm median move, well inside the 8 cm path-radius guard. Computed with `deploy/tools/workspace_spec.py`.

VI. A GUARDED NEEDLE-HANDLING SEQUENCE

A. Phase machine

Fig. 5 shows the sequence. The task is specified by three *tool* poses in the frame of `measured_cp`: the start pose, read from the arm; the grasp pose; and the suture pose, at which the needle sits angled at the entry point. Needle-frame targets cannot be used, because after a blind grasp the needle-in-jaw transform is unknown. The approach leg may use the RL policy or a staged SE(3) servo (D2). Transport uses the servo. The Place policy runs in shadow mode only: on the recorded geometry it reports itself out of distribution as soon as transport begins, which is consistent with its replay results above.

No grasp sensor. The simulator attaches the needle by fiat, and hardware has no equivalent. Grasp evidence comes from two indirect channels: the jaw-angle residual relative to an empty-jaw baseline, and jaw effort, which not every arm publishes. Both are logged with `grasp_verified=false`. The default gate is manual. Jaw units are a further sim-to-real trap: on the dVRK negative jaw angles mean squeeze, so mapping “closed” to 0 applies no tendon tension at all. The grip is therefore commanded on a separate channel (default

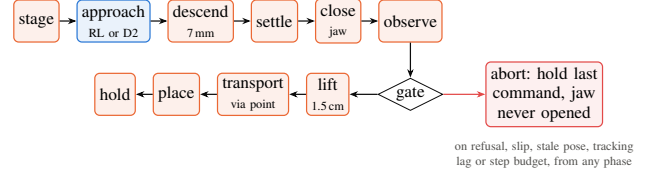


Fig. 5. The real-robot phase machine. Staging places the arm inside the policy’s support, and *descend* covers the 7 mm standoff that training never covers. The gate is either a human confirmation (default) or jaw evidence. Transport goes through a via point one lift height above the suture pose, so the needle cannot pass below the tissue plane. The suture pose is corrected for where the jaws actually closed.

−15°).

Suture-pose compensation. A suture pose taught by jogging the arm while it holds the needle encodes how the needle sat in the jaws at that moment. Let W be the needle pose, E its required pose and G the grasp. The tool must reach $S = E W^{-1} G$. A pose S_t taught with grasp G_t therefore transfers to an actual grasp G_a as

$$S_a = S_t G_t^{-1} G_a, \quad (3)$$

in which neither E nor W appears. Corrections larger than 10 mm abort, because a correction that large most likely means the needle moved.

Safety envelope. A fail-closed precheck verifies reachability, the workspace and operator boxes, step budgets, and tolerances against the lift distance and the deadband. The operator must state the lift sign, because $+z$ in the ECM frame is not guaranteed to point away from the tissue. Runtime guards cap each command at 2.5 mm and 5°, abort on 1.5 cm of tracking lag or on a `measured_cp` older than 0.25 s, and never open the jaw on abort.

B. Hardware bring-up findings

Position deadband. During the first live staging move, the arm stayed at 0.25 cm error for 49 cycles without moving at all. A single hand-published 2 mm setpoint moved it 1.947 mm. The loop, however, was publishing about 0.4 mm per cycle, because a proportional servo shrinks its step near the goal, and the arm ignored these commands. No existing guard detected this, since the command never ran more than 0.4 mm ahead of the arm. We added a minimum command

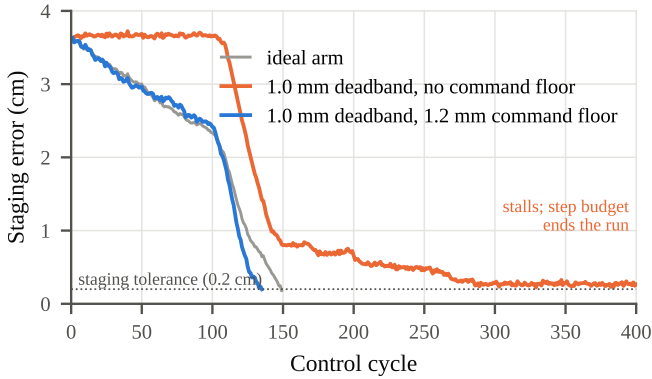


Fig. 6. Staging under a 1 mm position deadband (offline, arm lag 0.5, 0.2 mm pose noise). Without a command floor, the servo’s shrinking steps are absorbed and the move stalls, as on the hardware, until the step budget ends the run. A 1.2 mm floor restores convergence, and the full nine-phase sequence completes 0.18 cm from the suture pose.

TABLE III

OFFLINE REHEARSAL OF THE FULL SEQUENCE ON THE RECORDED TASK
(D2 APPROACH, 7 MM STANDOFF, 0.2 MM NOISE, SEED 1).

Arm lag per cycle	0.0	0.5	0.7
Cycles to completion	166	241	337
Final error to suture pose (cm)	0.064	0.076	0.098
Precheck	pass	pass	pass

magnitude that stretches a non-zero command up to the measured deadband, along the same direction and never past the target. The precheck now rejects any tolerance finer than the deadband. Fig. 6 reproduces the stall with a deadband model and shows the fix.

Middleware. State topics on the dVRK do not all use the same QoS. A RELIABLE subscriber matched `measured_cp` but not `jaw/measured_js`, so the jaw appeared absent even though `ros2 topic echo` showed it publishing. All subscriptions now default to BEST_EFFORT. The node also refuses to execute when nothing is subscribed to its command topic, because publishing into the void looks identical to an arm that does not move.

Operator interface. On the first dry run, the gate prompt printed once and then scrolled away under a 10 Hz status line, so the paused arm looked hung. The prompt now repeats and status output is throttled while waiting. A keyword-substitution error that reached the node’s `untested main()` led to static checks. Every keyword and every parsed argument is now checked against the real signatures using the Python AST. The deployment package has 588 unit tests, none of which require ROS, a robot or a checkpoint.

C. Offline rehearsal of the full sequence

Every change is rehearsed against a kinematic arm model with first-order lag, pose noise, a jaw model and an optional deadband, using the recorded operator poses. Table III and Fig. 7 show the full sequence with the D2 approach controller and the 7 mm standoff.

VII. VISION-GUIDED GRASP CALIBRATION

A. Two stages

The coarse stage maps a FoundationPose estimate through the hand-eye transform to a grasp target $p_{\text{nom}} = {}^E\mathbf{T}_C {}^C\mathbf{T}_N \mathbf{t}_{\text{grasp}}$. The fine stage learns the residual $r = p_{\text{grasp}} - p_{\text{nom}}$ from grasps taught by hand, following Xiao *et al.* [13], with two departures. First, we use trivariate polynomials of *total* degree n in the Bernstein basis. The constant and affine baselines are then the degree-0 and degree-1 rungs of the same ladder. The model is frame-invariant and needs 1/4/10/20 coefficients rather than 1/8/27/64 for the tensor-product basis. Second, the penalty is the exact thin-plate curvature energy, whose null space is the affine functions. As $\lambda \rightarrow \infty$ the fit therefore becomes the affine model, and choosing λ by leave-one-placement-out cross-validation decides whether curvature has earned its place. The fit is linear and is solved with one QR factorization. The convex-hull property bounds the correction anywhere in the calibrated box, and the runtime refuses to extrapolate outside it.

B. Predicted residual structure

Before collecting any data, we propagate each error source through the dVRK DH parameters over a 4 cm grasping box (Fig. 8a). Hand-eye error is *exactly* affine: $r = (R_{\text{true}} R_{\text{nom}}^\top - I)p_{\text{nom}} + b$. Offsets in the first two PSM joints rotate the whole distal chain about axes through the base, so they are also affine. Every other geometric term leaves less than 0.02 mm after an affine fit, about twenty times below the repeatability of a taught grasp. A positive result at degree ≥ 2 would therefore be a statement about FoundationPose’s learned bias, not about the robot. Two choices matter more than the degree. The needle mesh origin is the centre of the arc, 10.18 mm from the wire. Using it instead of the SurgicAI grasp point on the arc raises the cross-validated error from 0.46 to 3.07 mm at $\pm 30^\circ$ yaw. In addition, the noise floor is correct only when the taught grasp itself is repeated from a fresh approach. With one grasp per placement the floor reads 0.31 mm, while the achieved error is 0.64 mm. In a power analysis with a 1 mm curved bias, 45 placements detected curvature in 6/6 seeds and produced no false positives at any sample size. We therefore plan 60 placements.

C. First hardware samples

Ten placements have been recorded on our dVRK (Fig. 8b), and none can yet be used in a fit. *First*, the gripper poses were judged “ready to grasp” by eye, and the jaws were not closed on the needle. The per-axis correlation between the needle and gripper positions is 0.64, 0.76 and 0.15 for x , y and z , the signature of an operator reading depth from a flat endoscope image. *Second*, the residual contains a constant of about 43 mm, mostly in y . A constant offset is invisible to a transform-invariant check, since rigid transforms preserve distances. Our leading hypothesis is a frame mismatch: `measured_cp` is published in an “ECM” frame, while the hand-eye transform maps the camera to the ECM *tip*. Consecutive samples with the wrist held steady

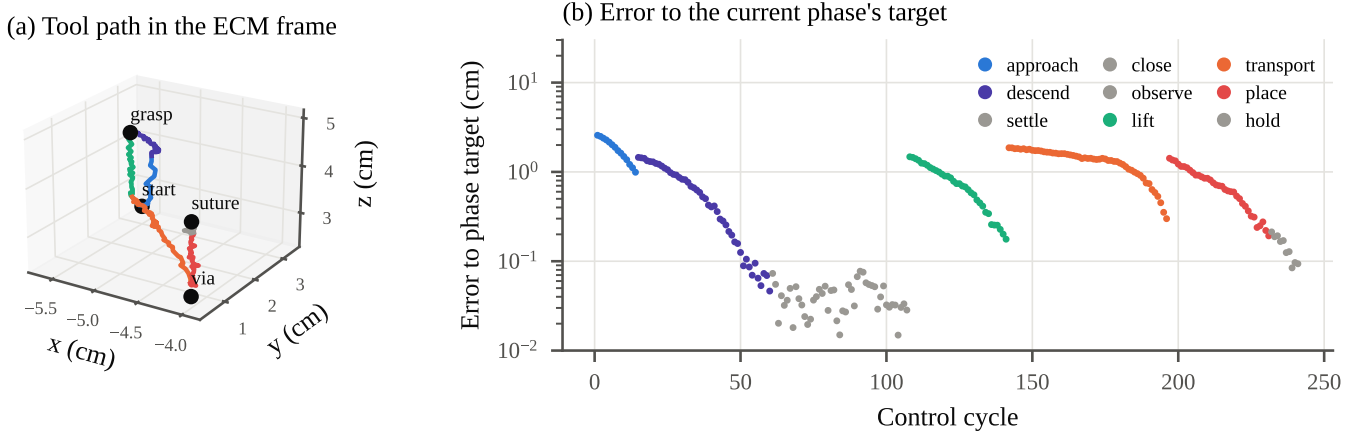


Fig. 7. One rehearsal of the full sequence (arm lag 0.5). (a) Measured tool path in the ECM frame, colored by phase. (b) Error to each phase’s own target. Every motion phase converges before the next begins, and the jaw phases (grey) hold position while the gate is evaluated.

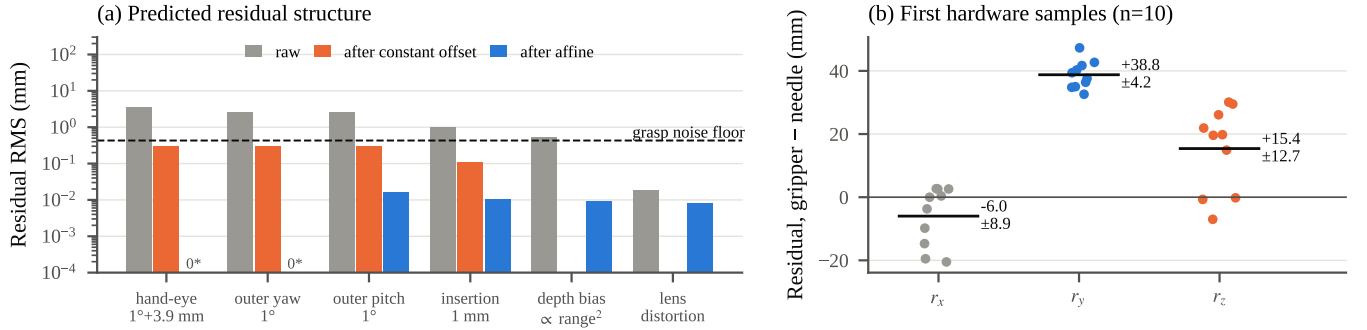


Fig. 8. (a) Residual RMS over a 4 cm box predicted from the dVRK kinematics for each error source: raw, after removing a constant, and after an affine fit. The dashed line is the rehearsed grasp noise floor, and 0* marks terms that are exactly affine ($< 10^{-12}$ mm). (b) Residuals of the first ten hardware samples. The near-constant $r_y \approx +39$ mm (s.d. 4.2 mm, over five days and $\sim 150^\circ$ of wrist variation) points to a frame-definition offset rather than to calibration error.

are consistent to within 1.2–1.8 mm, while samples taken minutes or days apart are not. The next session will confirm the frame and collect verified grasps: jaws closed, needle lifted 5–10 mm, and three fresh approaches per placement.

VIII. DISCUSSION AND LIMITATIONS

The largest obstacles to transfer were not dynamics or appearance. They were conventions: an Euler branch, a step size that exists in two versions, whether the observation closes the loop, a 7 mm standoff, and the sign of the jaw angle. Each one looks correct in isolation, and each one fails silently. Our main methodological recommendation is to measure a deployment contract by replaying the policy’s own demonstrations before running it on a robot, and to treat any mismatch between replay and the published rate as a bug.

This draft has clear limits. (i) No closed-loop grasp or suture on the real robot has been completed yet. The hardware results are bring-up findings, dry runs and calibration samples, and the complete sequence has been validated only offline. (ii) The hardware RL-versus-servo comparisons made before the contract fixes are void, and they have not yet been repeated. (iii) The high-level PPO checkpoint is not available, so the hierarchy results use scripted sequencing, and $n=20$ gives wide intervals. (iv) Place reproduces only

12/25 at a 10° tolerance and does not drive the arm. (v) FoundationPose tracking of the PSM is not yet reliable (Sec. IV), and the needle-yaw cap of the pose audit limits the workspace. (vi) Hysteresis is not a function of position and cannot be represented by any $f(x, y, z)$. Keeping the approach direction consistent is the only mitigation we apply.

IX. CONCLUSION AND FUTURE WORK

We replicated the SurgicAI hierarchy on a modern ROS 2 stack and built a perception front end for the real endoscope. We then made the released policies executable on a real dVRK PSM through a verified contract, a staging solve and a guarded phase machine. Next steps, in order: resolve the 43 mm frame offset and collect verified grasps; run the empty-gripper rehearsal and then live grasp, lift and transport with the manual gate; repeat the RL-versus-servo comparison through the corrected contract; and retrain with start-pose randomization, which the released Approach demonstrations lack entirely (one distinct start pose across 50 episodes), and with a Place policy trained to a tight orientation tolerance.

APPENDIX

Reproducibility. The hierarchy replication is on branch jin-hierarchical-repro

of `xiangruiSun/SurgicAI` (`RL/run_jin_hierarchical.py`). The deployment, calibration and all offline tools are in the project repository under `deploy/`, with the simulation environments under `src/SurgicAI/RL/`. The replay numbers come from `tools/replay_demos.py`, the staging analysis from `tools/workspace_spec.py`, the residual structure from `tools/residual_structure.py` and the rehearsals from `tools/offline_grasp_lift.py`. The script that regenerates every data figure in this draft is in `paper/make_figures.py`.

Checkpoint registry. Upstream Approach TD3+HER+BC (SHA-256 prefix `06fc7813f93f`), upstream Place (`4fc615f07415`) and R6 (`6286a88c21f0`). All three act at $1.0\text{mm}/3^\circ$.

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